

ZAC STRYKER

AI APPLICATION DEVELOPER
AND LLM SOLUTIONS ENGINEER

EMAIL

zac@zacstryker.com

WEBSITE

<https://www.zacstryker.com/>

LINKEDIN

<https://www.linkedin.com/in/zacstryker-89333398/>

LOCATION

Utrecht, Netherlands

PROJECTS

Dungeons & Dragons Retrieval-Augmented Generation

End-to-end RAG chatbot performing semantic search over D&D 5e PDFs and synthesizing cited answers via the Claude LLM API, demonstrating enterprise RAG pipeline design.

Python · RAG · Vector Search · LLM APIs · sqlite-vec · pypdf

Text Summarization

Full-stack web app transforming long texts into concise summaries via four AI algorithms (extractive and abstractive) with compression metrics and file upload support.

Python · Flask · NLP · NLTK · Sumy · TextRank · LSA · Jinja2

Customer Segmentation

Interactive K-means++ clustering visualization with RFM analysis, animated scatter plots, silhouette scores, and elbow method for cluster quality evaluation.

Python · scikit-learn · pandas · Matplotlib · Seaborn · Chart.js

Mushroom Classification

Binary classification pipeline on the UCI Mushroom dataset with model toggling, GridSearchCV, parallel EDA plot loading, and structured analytical output.

Python · scikit-learn · pandas · Flask · Seaborn · REST APIs

PROFILE

AI Application Developer with 8+ years of enterprise experience across government, healthcare, finance, and legal sectors, specializing in LLM-powered applications, RAG pipelines, and semantic search. Built end-to-end AI solutions using Python, vector search, and LLM APIs, and engineered data pipelines processing 100M+ records with strict compliance requirements. Proven track record translating complex business requirements into scalable backend services, structured APIs, and governance-aligned data workflows.

WORK EXPERIENCE

Day Pitney LLP

REMOTE
Sep 2023 - Present

FINANCIAL SYSTEMS SUPPORT SPECIALIST

- Developed AI-assisted web applications and automated cloud solutions for finance operations, reducing manual processing time by 75% across 600+ attorneys.
- Migrated 6 legacy SSIS packages to modern Azure-based workflows, cutting pipeline maintenance overhead by 50% and improving reliability.
- Translated complex business logic into robust SQL and Python database solutions, supporting 20+ finance workflows with full documentation.
- Led cross-functional delivery projects across 3 departments, ensuring on-time completion with compliance-aligned, governance-ready solutions.

United States Defense Health Agency

REMOTE
Dec 2021 - Jul 2023

BUSINESS INTELLIGENCE DEVELOPER

- Engineered ETL pipelines processing 100M+ records from 4 source systems, improving overall data quality by 60% across enterprise healthcare reporting.
- Built 12+ Tableau and QlikView dashboards tracking healthcare quality KPIs sourced from 9.5M+ beneficiaries and 6K+ providers.
- Implemented HIPAA-compliant data anonymization safeguards across 5 reports, ensuring regulatory compliance for sensitive patient data.
- Administered enterprise reporting environments including security configuration and automated refresh scheduling for 6K+ provider users.
- Delivered compliance and patient care dashboards that saved end users 5 hours per week through streamlined, self-service analytics.

Florida Department of Health

TALLAHASSEE, FL, USA
Oct 2020 - Dec 2021

TECHNICAL SUPPORT ANALYST

- Engineered COVID-19 data pipelines processing 3M+ daily records across all 67 Florida counties, delivering analytics to support public health decision-making.
- Optimized QVD/QVF extracts and REST/SQL integrations, improving Qlik report load times by 70% through systematic pipeline tuning.
- Designed a KPI tracking model covering 5 key metrics for the Support department, driving an 80% improvement in response efficiency.
- Built Power BI Sprint Capacity reports tracking velocity for 3 agile teams, improving sprint planning consistency by 25%.

SKILLS

AI / LLM DEVELOPMENT

- > RAG
- > LLM APIs
- > Vector Search
- > Prompt Engineering
- > sentence-transformers

MACHINE LEARNING / NLP

- > Transformers
- > NLP
- > scikit-learn
- > TensorFlow
- > Neural Networks

PYTHON / BACKEND

- > Python
- > Flask
- > REST APIs
- > Server-Sent Events
- > Jinja2

DATA / SQL

- > SQL/T-SQL
- > ETL
- > Snowflake
- > pandas
- > Data Modeling

CLOUD / INFRASTRUCTURE

- > Microsoft Azure
- > Docker
- > NGINX
- > Ubuntu
- > Power Automate

WORK EXPERIENCE

CONTINUED

Aderant

TALLAHASSEE, FL, USA
Jun 2018 - Apr 2020

INTERMEDIATE TECHNICAL SUPPORT ANALYST

- Advised 20+ enterprise law firm clients on deployment strategies and infrastructure design, reducing deployment incidents by 60%.
- Optimized database performance by up to 75% for high-load environments through systematic diagnostic tuning and SQL query analysis.
- Executed environment deployments, patches, and custom configurations for 2K+ enterprise law firms with 100% on-time delivery.
- Resolved complex connectivity and performance bottlenecks, cutting average resolution time by 30% across the client portfolio.

FSU College of Social Work

TALLAHASSEE, FL, USA
Dec 2017 - Jun 2018

TECHNICAL SUPPORT ANALYST

- Served as Helpdesk Technician and interim IT Manager for a research institution with 1,200 students.
- Managed physical devices, networking, and domain access for 60+ faculty and staff.
- Supported day-to-day IT operations ensuring uptime and reliable access across the institution.

First Commerce Credit Union

TALLAHASSEE, FL, USA
Sep 2016 - Nov 2017

DIGITAL SUPPORT TECHNICIAN

- Supported 140+ staff across a dozen locations during a large-scale migration to VMware virtual desktops, ensuring minimal disruption.
- Constructed an extensive SharePoint knowledge base to standardize support call procedures across all front-line technicians.
- Represented IT and eServices departments in Process Improvement Program meetings, providing guidance on inefficient operational practices.

EDUCATION

Florida State University - College of Music

Bachelor of the Arts, Music